Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Agent Security

Course Code: C28

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	First edition.	Dr Alfred Ang
2.0	7 November 2026	Rebuilt from the parent course's courseware v5.0: Claude Code theme, Topic 1 and Lab 1 from the Human-AI Workforce course, six labs on prompt injection, jailbreak, data leak, an AI Governance Policy and an AI Incident Report, each with mock data, scripts and a PDF of prompts. Day retimed to 9:30am to 5:30pm, 7.5 instructional hours.	Dr Alfred Ang

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TABLE OF CONTENTS

Course Information.....4
Learning Outcomes.....4
Course Schedule.....4
Lab Reference.....5
Resources.....5

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Course Information

Item	Detail
Course Title	AI Agent Security
Course Code	C28
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Duration	1 day, 7.5 instructional hours
Daily Timing	9:30 am to 5:30 pm, with one 30-minute lunch
Mode	Classroom Facilitated Training. Learners bring a laptop
Course scenario	Kopi Kaki Co., a fictional Singapore kopi startup. All lab data is fictional
Slide deck	AI Agent Security (C28)-v2.0.pptx (121 slides)
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
- LO2: Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
- LO3: Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

Course Schedule

Time	Duration	Topic / Activity	Slides
9:30–9:40	10 min	Welcome, trainer, ground rules, course outline and lab set-up	1–9
9:40–10:25	45 min	Topic 1: From rules to learning to action; four engineering lenses 2023 to 2026; agentic AI versus AI agents; how an agent works like a person; the agentic loop; how generative AI writes; generative versus discriminative models; AI does the work, humans decide	10–21
10:25–11:00	35 min	Topic 1: Five Human-AI collaboration scenarios; use cases; AI for science; OpenClaw to Hermes to hosted always-on agents; the Big Five always-on agents compared	22–41
11:00–11:35	35 min	Lab 1: Your always-on CEO assistant. Screenshot for your lab record	42–49
11:35–12:10	35 min	Topic 2: Prompt engineering and data quality; definition and examples of prompt injection; why a model cannot tell a genuine prompt from a malicious one; consequences	50–59
12:10–12:45	35 min	Lab 2: Install Hermes Desktop on a free model, simulate prompt injection, fix it. Screenshots for your lab record	60–67
12:45–13:00	15 min	Topic 2: Definition and examples of jailbreak; techniques; what stronger models and guardrails change	68–72
13:00–13:15	15 min	Lab 3: Simulate a jailbreak, fix it. Screenshots for your lab record	73–78
13:15–13:45	30 min	Lunch	-
13:45–14:00	15 min	Topic 2: Definition and examples of data leak; personal data and the PDPA	79–82
14:00–14:15	15 min	Lab 4: Simulate a data leak, fix it. Screenshots for your lab record	83–89
14:15–15:40	85 min	Topic 3: How a model works; limits, bias and ethics; the five principles of AI security explained, with examples	90–105
15:40–15:50	10 min	Topic 3: The frameworks behind the five principles: IMDA, NIST, OWASP, ISO, PDPA	106
15:50–16:35	45 min	Lab 5: Create an AI Governance Policy from an interview prompt.	107–111

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		Screenshot for your lab record	
16:35–16:45	10 min	Topic 3: Incident response under the PDPA: Contain, Assess, Report, Evaluate	112
16:45–17:30	45 min	Lab 6: Create an AI Incident Report from mock agent traces. Screenshot for your lab record	113–118

Lab Reference

Lab	Title	Slides
Lab 1	Your Always-On CEO Assistant: Meta Muse or OpenAI Dots	42–49
Lab 2	Prompt Injection: One Fake Note in a Customer Review	60–67
Lab 3	Jailbreak: 'It's Only a Comedy Skit'	73–78
Lab 4	Data Leak: 'Just Send the Agency a Sample'	83–89
Lab 5	Create an AI Governance Policy: Your Agent Interviews You	107–111
Lab 6	Create an AI Incident Report: The Leak From Three Months Ago	113–118

Lab materials: one folder per lab under labs/, holding LAB-0N-Instructions.pdf, data/ and scripts/. Prompts can be copied from prompts/index.html, also published at https://tertiarycourses.github.io/C28-AI-Agent-Security/prompts/index.html.

Resources

- Learner laptop (Mac on Apple Silicon, or Windows) with Hermes Desktop
- Free Nous Portal account; a ChatGPT plan for the stronger model where available; Meta Muse account
- Training Google account seeded with the Lab 1 mock meetings and emails, prepared by the trainer
- Slide deck, Learner Guide, lab PDFs and the copy-prompt page

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